become equal and opposite the applied voltage. The charged capacitor is now full of stored energy. This stored energy can be used again when needed.

The factors on which the capacity of capacitor depends are:

a. Directly proportional to the area of the plate of the condenser.
b. Inversely proportional to the distance between the plate.
c. The nature of di-electric insulating material.

### 2.7.2 Capacitance(C)

Capacitance of capacitor is defined as the ratio between the charge given to the condenser and supply voltage. It is denoted by the letter C.

Suppose a charge +ve 'q' units is given to condenser, the potential will then be raised. The greater the charge given to the condenser, the greater is the rise in its potential.

we have $q \alpha v$ (or) $\frac{q}{v}$ constant.

$C = \frac{q}{v}$ farad

Hence, c is the capacity of the condenser.

q is the charge given to the condenser

v potential difference between the plates.

The unit of capacity is 'Farad'. A capacitor is said to posses a capacity of one farad, when its potential is raised by one volt, and when one coulomb of charge is given to it.

$$1 \text{ Farad} = \frac{1 \text{ coloumb}}{1 \text{ volt}}$$

$2\pi \times 10^{18}$ number of electrons is mentioned as one coulomb.

### 2.7.3 Types Of Capacitor

Usually there are three types of capacitor. They are

1) Fixed capacitor
2) Variable capacitor and
3) Electrolytic capacitor



**Figure 2.10** *Capacitors*

**i) Fixed Capacitor**

a) Mica Capacitor

Mica capacitor is widely used in ratio circuit where fixed value condensers are required. These have metal foil sheets forming the coating and separated by a flat mica sheet as the dielectric medium

b) Paper Capacitor

Paper is rolled in the form of cylinder and dipper in wax solution in order to exhaust the air placed in between two thin aluminium plates. This type of capacitor is used in de-coupling circuits.

c) Ceramic Capacitor

These are the modern capacitors. In this, ceramic is used as di-electric



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